Campus Event Explorer

BIM493 - Mobile Programming I | Assignment #1

Assigned: October 13, 2026 | Due: October 25, 2026, 23:59 (UTC+3)
Duration: 12 days | Team: two students | Total: 100 points

Your task

Develop a Flutter mobile application that helps students discover campus events, save favorites and manage their own participation. Use fictional data stored in your Dart code. The app must work without an online service after installation.

1. Event data and Explore screen

- Create at least six events across at least three categories (for example Technology, Arts and Sports). Each event must have a unique ID, title, category, date/time, venue, description, local image, total capacity and initial available places.
- Include at least one initially full event and one event with exactly one available place. Use plausible dates after the assignment deadline. Keep available places between zero and total capacity.
- Display scrollable event cards with an image, title, category, date/time, venue, availability label, favorite control and a way to open the details. Use at least three distinct local images; category images may be reused.

2. Search and category filter

- Provide a title search that ignores letter case and surrounding spaces. Provide All and at least three category options. Search and category selection must work together; display the number of matching events.
- Show a readable empty-state message when no events match. Provide a way to clear the search. The same search/filter settings apply to Explore, Favorites and My Events.

3. Details and image viewer

- Open a separate event details screen showing all event information. A double-tap on its image must open another screen with an enlarged image. Provide normal Back navigation on both screens.
- Details must include a favorite control and a participation button. Returning to the previous screen must show the updated state.

Interaction requirements

4. Favorites and My Events

- Provide three views: Explore, Favorites and My Events. A bottom navigation bar or tabs are acceptable. Favorites contains only favorited events; My Events contains only joined events. Both must show an empty-state message when appropriate.
- Adding/removing a favorite from a card or details must update all relevant views during the same app session. Favorites do not change participation or capacity.

5. Join and cancel participation

- Joining must open a confirmation dialog showing the event title, date/time and venue, with Yes and No options. No or dismissing the dialog leaves the data unchanged. Yes joins the event once, decreases available places by one and shows feedback.
- A joined event must offer Cancel participation with confirmation. Confirming cancellation removes it from My Events and restores one available place. Joining the same event repeatedly must not create duplicates or decrease capacity twice.
- An unjoined event with zero available places must be marked Full and cannot be joined. A student who joined the last available place must still be able to cancel. Show the total number of joined events.

6. Layout, assets and font

- Use MaterialApp, Scaffold and appropriate layout widgets such as Row, Column, Padding and Expanded. Include at least one reusable StatelessWidget and one StatefulWidget. Use simple setState-based updates; advanced state-management libraries are not required.
- Declare your images and at least one custom font in pubspec.yaml and apply the font visibly. Bundle a local font or use google_fonts with locally bundled files. Credit images/fonts and include licenses where redistribution requires them.

Assessment (100 points)

Criterion	Points
Event data and Explore cards	15
Combined search, filters and empty states	15
Details, navigation and image enlargement	15
Favorites and consistency across views	15
Join/cancel logic, capacity and feedback	20
Layout, bundled images and custom font	10
Source organization, README and evidence	10

Scope and submission

Required Flutter version

Use Flutter 3.47.6 on the Stable channel for this assignment. Check your installed version using flutter --version and include its output in your README. Do not use the Beta or Main channel.

Scope and research guidance

Keep favorites and participation in memory during one app session. They may reset after restarting the application. Login, backend services, databases, real bookings, GPS and real phone calls are outside the required scope. Extra features do not compensate for missing required features and carry no bonus points.

Weeks 1-4 cover setup, widget composition, state concepts, layout, images, fonts and pubspec. Simple screen navigation, input handling, setState, dialogs and list filtering require a small guided research step. Start with the official Flutter recipes for navigation and fonts and the State.setState API documentation. Work on one interaction at a time.

What your submission must contain

- Submit your repository/archive link through ESTUOYS before the deadline. For private file hosting, enable link access so the instructor can download the files.
- Include lib/, pubspec.yaml, bundled assets and the platform/configuration files needed to run your project. Exclude build/ and .dart_tool/. Include pubspec.lock if generated. An APK is optional; source code is required.
- README: both full names and student IDs, each member's contribution, setup/run commands, asset credits and any known limitations. Add at least four screenshots covering Explore, details, Favorites and My Events, plus a short checklist showing that full-event, cancel-dialog and empty-search cases were checked.

Submitting the assignment:

- You would create a GitHub account and submit your files to be publicly available to the community or minimize your Project and upload to a file hosting service like Google Drive or Microsoft OneDrive etc. if you do not want to share your code with others.
- Late submission is not allowed by the system. Submissions via e-mail will not be accepted by the instructors and they will be deleted immediately.
- You should form groups of two and submit your assignment together. Your assignment file must include your student ID numbers, and you must put a readme file containing your full name and student ID number. I regret to inform you that those who do not have a readme file cannot claim rights to the assignment grades.

Due Date:

- October 25, 2026, Sunday midnight (23:59), Türkiye time (UTC+3). Late submission is not allowed by the system. Submissions via e-mail will not be accepted by the instructors and they will be deleted immediately.